

AutoMate – Campus Transport Coordination Application

Executive Summary | Project Proposal

Indian Institute of Information Technology (IIIT) Nagpur

Team Member Name	Student ID
Vivek Mishra	BT24CSE046
Gaurav Khambhat	BT24CSE050
Pratik Swami	BT24CSE051
Mantray Balsaraf	BT24CSE056

1. Introduction

AutoMate is a dedicated transport coordination mobile application designed specifically for IIIT Nagpur students and college-assigned electric auto-rickshaw drivers. The app addresses the chronic problem of unreliable, unstructured, and inefficient campus-to-main-road transport by introducing a digital queue-based scheduling system inspired by ride-hailing platforms like Ola/Uber — but adapted for a fixed-route, institutional context.

Key Features

- ✓ Queue-based time-slot scheduling system (20-minute slots, 9 AM–6 PM)
- ✓ Two directional queues: College → Main Road and Main Road → College
- ✓ Student and Driver separate login portals (college credentials)
- ✓ Automatic driver notifications on queue join
- ✓ Crowd management with FCFS (First-Come-First-Served) system
- ✓ Attendance/presence verification system by driver
- ✓ Schedule display with driver assignments

Target Users

Students Primary users accessing transport	Drivers Operating Electric Autos	Administration Oversight & Monitoring
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2. Problem Statement

2.1 Core Challenges

- **Low bus frequency:** Once per hour frequency forces students to wait long periods.
- **Unreliable information:** Auto-rickshaw drivers often give vague availability details.
- **Location mismatch:** Students near the main road can't reach campus when the auto is stationed at the college end.
- **Lack of structure:** No organized system to call, track, or book autos.
- **Privacy issues:** Multiple driver phone numbers circulate, frequently changing, leading to confusion.
- **Disturbance:** Ex-drivers continue to receive calls from students.
- **Chaos:** No organized crowd management when demand peaks.

2.2 Impact

Students miss classes or appointments due to delays.	Frustration and significant time loss for both parties.
Inefficient utilization of available electric autos.	Complete lack of accountability in the current manual system.

3. Objectives

3.1 Primary Objectives

1. Eliminate reliance on informal phone calls for auto coordination.
2. Implement a fair, transparent digital queue system.
3. Enable real-time driver notifications for demand.

- 4. Provide structured time-slot based scheduling (20-min slots).
- 5. Ensure attendance accountability with the tick/cross mechanism.

3.2 Secondary Objectives

- Provide students with live queue status visibility.
- Allow students to file verified complaints against absent drivers.
- Enable administration to monitor system usage and driver performance.
- Prevent misuse of the queue system via absence-penalty mechanism.

3.3 Long-term Vision

- Expand to include bus tracking and schedule notifications.
- Analytics dashboard for college administration.
- Predictive crowd management using usage history.
- Integration with college ERP/portal systems.

4. System Design & Technical Overview

4.1 User Types

Student Login

Authenticated via College Credentials

Driver Login

Admin-assigned dedicated accounts

4.2 Core Mechanism

The system revolves around **20-minute operational slots** running from 9:00 AM to 6:00 PM. Two digital queues are maintained simultaneously:

Queue 1: College → Main Road

Queue 2: Main Road → College

For each slot, the auto completes one round trip if either queue has students. The allocation is strictly **First-Come-First-Served (FCFS)** with a capacity of 4 passengers.

4.3 Attendance Handling

To ensure fairness, drivers verify student presence at the pickup point:

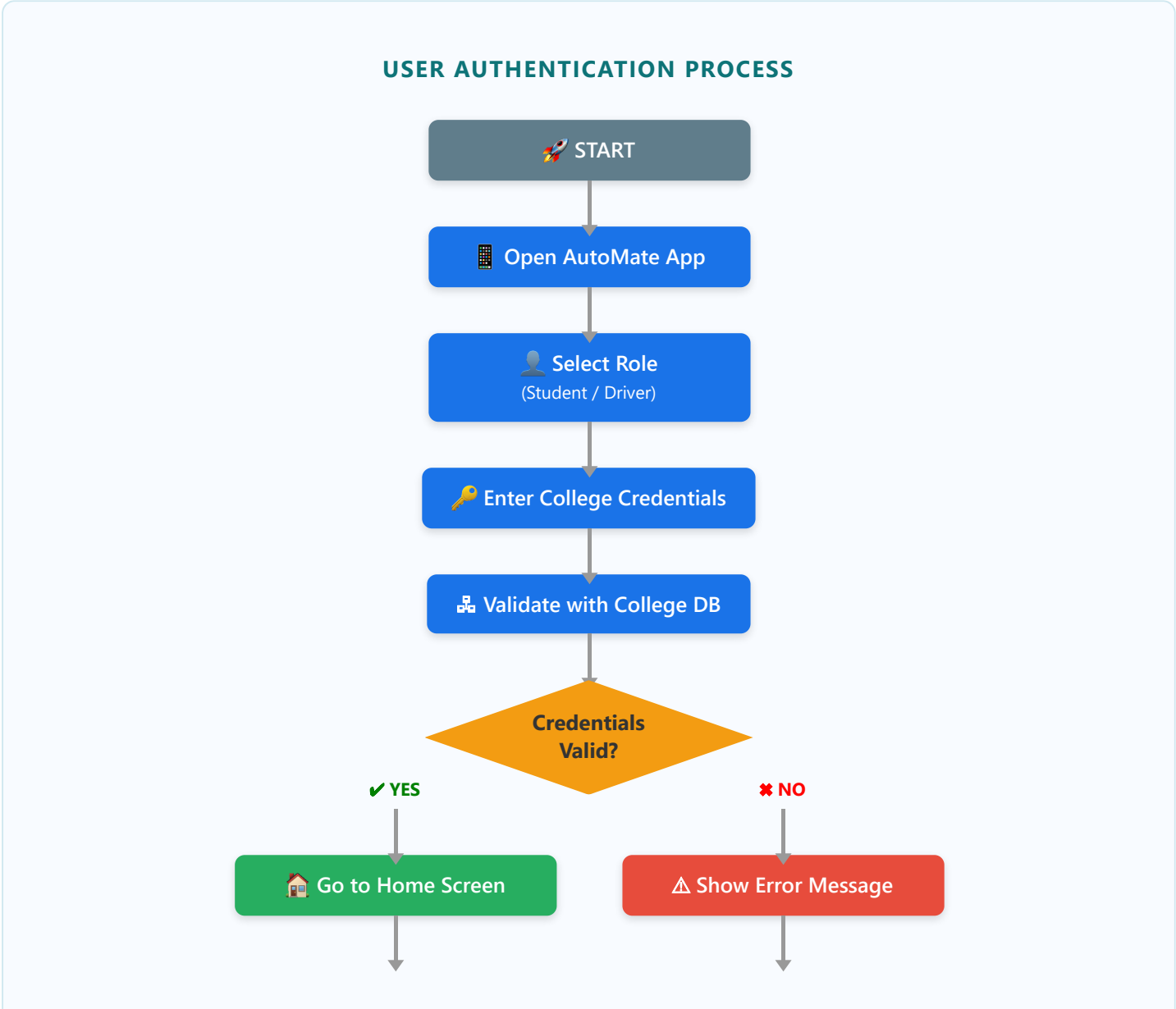
- ✔ **Present:** Student boards auto and is removed from the queue.
- ✖ **Absent:** Student is moved to the end of the queue. A second absence results in removal.

4.4 Tech Stack

Frontend	React Native / Flutter (Cross-platform)
Backend	Node.js with Express / Django (Python)
Database	Firebase Realtime DB / PostgreSQL
Authentication	College OAuth / JWT Tokens
Notifications	Firebase Cloud Messaging (FCM)
Hosting	AWS / Google Cloud

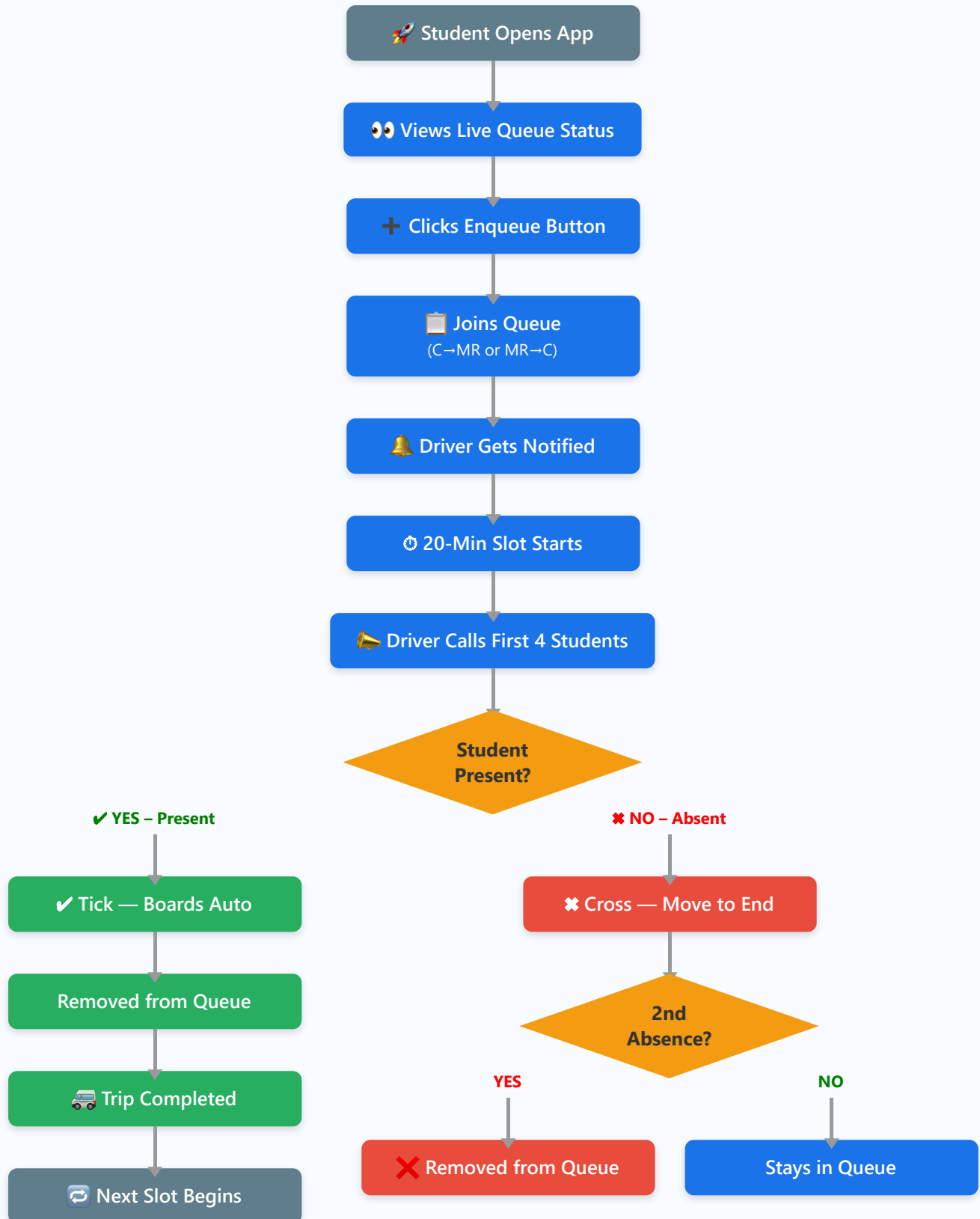
5. Process Flowcharts

Flowchart 1 — User Login Flow



Flowchart 2 — Queue & Trip Flow

STUDENT QUEUE & AUTO TRIP PROCESS



6. Competitive Analysis

Feature	AutoMate (Proposed)	Manual Phone Calls	Ola / Uber	WhatsApp Groups
Structured Queue	✓	✗	✓	✗
Driver Notification	✓	✓	✓	✗
Attendance Tracking	✓	✗	✗	✗
Complaint System	✓	✗	✓	✗
Institutional Integration	✓	✗	✗	✗
Crowd Management	✓	✗	✗	✗
No External Dependency	✓	✓	✗	✓

7. Project Timeline

Phase	Dates	Tasks
Phase 1	Mar 16 – 26	Requirements finalization, UI wireframes, DB schema, auth setup
Phase 2	Mar 27 – Apr 3	Queue logic implementation, student & driver interfaces
Phase 3	Mar 4 – Apr 10	Notification system, complaint module, attendance mechanism
Phase 4	Apr 10 – 18	Testing, bug fixes, admin dashboard, final deployment prep

8. Conclusion

8.1 Key Takeaways

AutoMate transforms an unreliable, ad-hoc transport system into a structured, transparent, and fair digital platform, ensuring equitable access to campus resources.

8.2 Expected Impact

70%+

Reduction in student wait-time
uncertainty

0

Ghost calls to off-duty or ex-drivers

100%

Accountability & transparency in
transport usage

8.3 Next Steps

Refer to the full **SRS (Software Requirements Specification)** document for detailed technical specifications, API definitions, database ERD diagrams, and high-fidelity UI mockups.

8.4 Final Statement

AutoMate is not just an app — it is a step toward a smarter, more connected, and efficient campus experience for IIIT Nagpur.



Implementation & Operational Workflow

This section summarizes how AutoMate will function in real-world campus operation and how the system coordinates students and drivers efficiently.

1. **Student Access** — *Students log in using their official college credentials. The application displays the current queue status, upcoming time slots, and estimated waiting time.*
2. **Joining the Queue** — *Students select their desired direction(College → Main Road or Main Road → College). The system places them in the digital queue using a First-Come-First-Served (FCFS) policy.*
3. **Driver Notification** — *When students join the queue, the assigned driver receives a real-time notification. The driver dashboard displays(Current queue, Upcoming passengers & Slot timing).*
4. **Trip Execution** — *At the beginning of each 20-minute slot, the driver calls the first four students in the queue. Students board the auto and the system updates automatically.*
5. **Attendance Verification** — **✓ Present** – Student boards the vehicle. **✗ Absent** – Student is moved to the end of the queue. If a student is marked absent twice, they are automatically removed from the queue to maintain fairness.
6. **System Monitoring** — *The administration can monitor - Queue activity, Trip completion rates, Driver availability, Student complaints.*



Operational Benefits :-

1. Fair Allocation of transport resources through structured queues.
2. Reduced waiting uncertainty for students.
3. Elimination of manual phone coordination.
4. Improved driver efficiency and accountability.
5. Administrative visibility into transport usage patterns.